

Comprehensive Technical Architecture, Methodology & Analysis Report

1. Executive Summary & Platform Overview

By synthesizing real-time vehicle sensor feeds, OEM databases, historical repair indices, and market depreciation curves through state-of-the-art Large Language Models (Anthropic Claude, Google Gemini, and xAI Grok), AAIA automates complex diagnostic workflows, maintenance forecasting, and financial total cost of ownership (TCO) modeling.

2.1 Frontend Presentation Tier

- **Security & Hardening:** Helmet HTTP protection headers, CORS origin whitelisting, bcryptjs password hashing, and

2.4 Data Base Search & Caching

- **Relational Storage (RAG):** Invector (extension for semantic vector embeddings across OEM repair manuals)
- **High-Speed Caching (Redis 7.4):** (for Redis) for high-speed VM decoding cache, session token blacklists, and rate-limiting
- **Vector Storage (RAG):** High-Speed Caching

- **Memory-Safe Client PDF Generation:** eliminating layout shifts and mobile browser crashes
- **Dynamic Database-Driven Pricing:** Decoupled billing pricing matrices from client code into PostgreSQL JSONB schemas,
- **Dynamic Database-Driven Pricing:** pricing modifications without application rebuilds
- **Mobile-First Responsive Retargeting:** implemented global CSS attribute overrides and container queries that dynamically
- **Mobile-First Responsive Retargeting:** single-column mobile views
- **Technical SEO & Semantic Structure:** engineered HTML5 semantic hierarchies, JSON-LD schemas, and Open Graph meta
- **Technical SEO & Semantic Structure:** proactive AI keywords.

4.1 Intelligent Diagnostics (SEV-0 to SEV-5)

4.2 5-Year Total Cost of Ownership (TCO) Analysis

4.3 Fleet Management & Health Indexing

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5. Technical Architecture Summary

Layer	Technologies / Specifications
Presentation Tier	React 18, React Router v6, Recharts, jsPDF, html2canvas, Tabler Icons
Server Gateway	Node.js 20+, Express.js, Socket.IO, Helmet, Morgan, Winston
AI Reasoning Layer	Anthropic Claude 3.5, Google Gemini 1.5/2.0, xAI Grok / OpenAI
Database & Vector	PostgreSQL 14+ with pgvector, Connection Pooling
Cache & Session	Redis 7+ (ioredis), node-cache memory fallback
Security & Auth	JWT (JSON Web Tokens), bcryptjs, RBAC, Rate-Limiting
Data Integrations	AutomotiveDataset.com, NHTSA Safety & Recalls API